U-5 Matrix Decomposition and their Applications.

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3:37.

Similar Matrices: Two $n \times n$ matrices A and B are similar if there is an invertible matrix P such that $B = P^{-1}AP$.

Diagonalization:

A square matrix is diagonalizable if it is similar to a diagonal matrix.

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Therefore, the square matrix A is diagonalizable if $A = PDP^{-1}$.
Consider $A = PDP^{-1} \Rightarrow AP = PD$ and $P = [p_1\ p_2\ p_3 \dots p_n]$
$\Rightarrow [Ap_1\ \ Ap_2\ \ Ap_3 \dots Ap_n] = [d_1p_1\ d_2p_2\ d_3p_3 \dots d_np_n]$
Hence $p_1, p_2, p_3, \dots, p_n$ are the eigenvectors of A and $d_1, d_2, d_3, \dots, d_n$ are the eigenvalues of A.
This implies that a square matrix is diagonalizable if and only if the matrix is non-defective i.e. (A has n linearly independent eigenvectors)

Sufficient condition for diagonalization: If an $n \times n$ matrix A has n distinct eigenvalues, then the corresponding eigenvectors are linearly independent and A is diagonalizable.

Note:
- Similar matrices have the same eigenvalues.

Working Rule to diagonalize a given square matrix A
1. Determine all the eigenvalues of A and find the corresponding linearly independent eigenvectors $p_1, p_2, p_3, \dots, p_n$.
2. Construct the modal matrix $P = [p_1\ p_2\ p_3 \dots p_n]$ and compute P^{-1}.
3. Compute diagonal matrix $D = P^{-1}AP = diag(\lambda_1, \lambda_2, \dots, \lambda_n)$ where each eigen value satisfies $Ap_i = \lambda_i p_i$.

This diagonalization can be applied to compute integral powers of the matrix A: i.e., $A^n = PD^nP^{-1}$.

I. find the modal matrix P which diagonalizes the following matrices.

1) $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ also find A^4

→ C. E. n's,

$$\lambda^2 - Tr(A)\lambda + |A| = 0$$

or, $\lambda^2 - 7\lambda + 10 = 0$

$\therefore \lambda = 5, 2$

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for $\lambda = 5$

$$A - \lambda I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$\approx \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

$n - r = 2 - 1 = 1$ free variable

So,

Let,

$y = k$

$\therefore \quad x = k$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore \quad P_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Similarly, for $\lambda = 2$

$$A - \lambda I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$$\approx \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$$

$\therefore \; n - r = 2 - 1 = 1$ free variable

Let,

$y = k$

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So,

$2x + y = 0$

$\therefore \; x = \frac{-k}{2}$

$$\therefore \quad X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k/2 \\ k \end{bmatrix} = \frac{-k}{2} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\therefore \quad P_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Thus, Modal Matrix is

$$P = [P_1 \; P_2]$$

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$$= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

Also,

$$P^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$ → Use calculator not your brain :)

So,

$$D = P^{-1} A P$$

$$= \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \cdot \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

Now, for A^4

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WKT,

$$A^n = P D^n P^{-1}$$

So,

$$A^4 = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}^4 \cdot \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 422 & 203 \\ 406 & 219 \end{bmatrix}$$

//

2. $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.

→ C.E. n's,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = -3$$
$$S_2 = 4 - 8 + 4 = 0$$
$$S_3 = 4$$

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Thus,

$$\lambda^3 + 3\lambda^2 - 4 = 0$$

$\therefore \lambda = 1, -2, -2$

for $\lambda = 1$

$$A - \lambda I = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$\approx \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 0 & -3 & -3 \end{bmatrix}$$

$R_3 \to R_3 + R_1$

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$$\approx \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 3 - 2 = 1$ free variable

Let $z = k$

$\therefore y = -k$

$-3x - 6y - 3z = 0$

$\therefore x = k$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

So, $P_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ //

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Similarly for $\lambda = -2$

$$A - \lambda I = \begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + R_1$

$R_3 \longrightarrow R_3 - R_1$

$$\approx \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 3 - 1 = 2$ free variable

Let, $z = k_1$, $y = k_2$

$\therefore x = -k_2 - k_1$

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$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k_2 - k_1 \\ k_2 \\ k_1 \end{bmatrix}$$

$$= k_2\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + k_1\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore P_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Thus, modal matrix P,

$P = [P_1 \; P_2 \; P_3]$

$$= \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

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Also,

$$P^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

Thus.

$D = P^{-1} A P$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

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3. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.

↳ Ch. eq's,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = -1$$

$$S_2 = -12 - 3 - 6 = -21$$

$$S_3 = 45$$

Thus,

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\therefore \lambda = 5, -3, -3$$

for $\lambda = 5$

$$A - \lambda I = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix}$$

$$R_1 \to R_1 - 7R_3$$

$$R_2 \to R_2 + 2R_3$$

$$\approx \begin{bmatrix} 0 & 16 & 32 \\ 0 & -8 & -16 \\ -1 & -2 & -5 \end{bmatrix}$$

$$R_1 \to R_1 + 2R_2$$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & -8 & -16 \\ -1 & -2 & -5 \end{bmatrix}$$

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$\therefore$ $n - r = 3 - 2 = 1$ free variable

Let,

$$z = k$$

So $y = -2k$

$\therefore x = -k$

$\therefore P_1 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

for $\lambda = -3$

$A - \lambda I = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 + R_1$

$\approx \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

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$\therefore n - r = 3 - 1 = 2$ free var

Let

$z = k_1, \; y = k_2$

$\therefore x = 3k_1 - 2k_2$

$\therefore X = \begin{bmatrix} 3k_1 - 2k_2 \\ k_2 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

$\therefore P_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \; P_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

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Thus

Modal matrix, $P = \begin{bmatrix} -1 & 3 & -2 \\ -2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$$P^{-1} = \begin{bmatrix} -1/8 & -1/4 & 3/8 \\ 1/8 & 1/4 & 5/8 \\ -1/4 & 1/2 & 3/4 \end{bmatrix}$$

now,

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -1/8 & -1/4 & 3/8 \\ 1/8 & 1/4 & 5/8 \\ -1/4 & 1/2 & 3/4 \end{bmatrix} \cdot \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \cdot \begin{bmatrix} -1 & 3 & -2 \\ -2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

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$$= \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} //$$

4. $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ also find A^6.

→ C.E. is,

$$\lambda^2 - Tr(A)\lambda + |A| = 0$$

or, $\lambda^2 - 8\lambda + 15 = 0$

$\therefore \lambda = 5, 3$

for $\lambda = 5$

$$A - \lambda I = \begin{bmatrix} 2 & 2 \\ -4 & -4 \end{bmatrix}$$

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$R_2 \to R_2 + 2R_1$

$$\approx \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$$

$\therefore n - r = 2 - 1 = 1$ free var

Let

$y = K$

$\therefore x = -K$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k \\ k \end{bmatrix} = -k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore P_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

for $\lambda = 3$

$$A - \lambda I = \begin{bmatrix} 4 & 2 \\ -4 & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$\approx \begin{bmatrix} 4 & 2 \\ 0 & 0 \end{bmatrix}$$

$\therefore n - r = 2 - 1 = 1$ free variable

Let,

$$y = k$$

$$\therefore x = \frac{-k}{2}$$

$$\therefore X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -k/2 \\ k \end{bmatrix} = \frac{-k}{2} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\therefore P_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Thus, modal matrix.

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$$

Also

$$P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

So,

$$D = P^{-1} A P$$

$$= \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} //$$

now, $A^6 = P \cdot D^6 \cdot P^{-1}$

$$= \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}^6 \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 30521 & 14896 \\ -29792 & -14167 \end{bmatrix}$$

5. $A = \begin{bmatrix} 0 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$

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C.E. is,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = 1$$
$$S_2 = -6 - 4 + 0 = -10$$
$$S_3 = 8$$

Thus,

$$\lambda^3 - \lambda^2 - 10\lambda - 8 = 0$$

$\therefore \lambda = 4, -1, -2$

for $\lambda = 4$

$$A - \lambda I = \begin{bmatrix} -4 & 0 & -2 \\ 0 & -6 & 0 \\ -2 & 0 & -1 \end{bmatrix}$$

$$R_3 \to 2R_3 - R_1$$

$$\approx \begin{bmatrix} -4 & 0 & -2 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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$\therefore n - r = 3 - 2 = 1$ free variable

Let,

$z = k$

$\therefore y = 0$

$\therefore x = \frac{-k}{2}$

$$\Rightarrow X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{k}{2} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$\therefore P_1 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

for $\lambda = -2$

$$A - \lambda I = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 5 \end{bmatrix}$$

$$R_3 \to R_3 + R_2$$

$$\approx \begin{bmatrix} 2 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$n - r = 3 - 2 = 1$ free variable

Let,

$z = k$

$\therefore y = 0$

$\therefore x = k$

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$$P_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

for $\lambda = -1$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & -2 \\ 0 & -1 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

$$R_3 \to R_3 + 2R_2$$

$$\approx \begin{bmatrix} 1 & 0 & -2 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 3 - 2 = 1$ free variable

Let,

$z = k$

$\therefore y = 0$

$\therefore x = 2k$

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$\therefore P_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

Thus, Modal Matrix

$$P = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

Also.

$$P^{-1} = \begin{bmatrix} -1/5 & 0 & 2/5 \\ 0 & 1 & 0 \\ 2/5 & 0 & 1/5 \end{bmatrix}$$

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Thus.

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -1/5 & 0 & 2/5 \\ 0 & 1 & 0 \\ 2/5 & 0 & 1/5 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} //$$

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6. $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ also find A^4.

C.E. is,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = 7$$

$$S_2 = 4 - 8 + 4 = 0$$

$$S_3 = -36$$

Then

$$\lambda^3 - 7\lambda^2 + 36 = 0$$

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$\therefore \lambda = -2, 6, 3$

for $\lambda = -2$

$$A - \lambda I = \begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix}$$

$R_1 \rightarrow R_1 - 3R_2$

$R_2 \rightarrow R_3 - 3R_2$

$$\approx \begin{bmatrix} 0 & -20 & 0 \\ 1 & 7 & 1 \\ 0 & -20 & 0 \end{bmatrix}$$

$R_3 \rightarrow R_3 + R_1$

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$$\approx \begin{bmatrix} 0 & -20 & 0 \\ 1 & 7 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 1$ free variable

Let,

$$z = k$$

$$\therefore y = 0$$

$$\therefore x = -k$$

$\therefore P_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

for $\lambda = 6$

$$A - \lambda I = \begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

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$R_1 \to R_1 + 5R_2$

$R_3 \to R_3 - 3R_2$

$$\approx \begin{bmatrix} 0 & -4 & 8 \\ 1 & -1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$$

$R_1 \to R_1 - 2R_3$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$$

$\therefore n - r = 3 - 2 = 1$ free var

Let,

$z = k$

$\therefore y = 2k$

$\therefore x = k$

$\therefore P_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

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for $\lambda = 3$

$$A - \lambda I = \begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix}$$

$R_1 \to R_1 + 2R_2$
$R_3 \to R_3 - 3R_2$

$$\approx \begin{bmatrix} 0 & 5 & 5 \\ 1 & 2 & 1 \\ 0 & -5 & -5 \end{bmatrix}$$

$R_3 \to R_3 + R_1$

$$\approx \begin{bmatrix} 0 & 5 & 5 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 1$ free

Let,

$z = k$

$\therefore y = -k$

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$\therefore x = k$

$$\therefore P_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Thus, Modal Matrix,

$$P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

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Also,

$$P^{-1} = \begin{bmatrix} -1/2 & 0 & 1/2 \\ 1/6 & 1/3 & 1/6 \\ 1/3 & -1/3 & 1/3 \end{bmatrix}$$

Thus,

$D = P^{-1} A P$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

now,

$$A^4 = P D^4 P^{-1}$$

$$= \begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}^4 \cdot \begin{bmatrix} -1/2 & 0 & 1/2 \\ 1/6 & 1/3 & 1/6 \\ 1/3 & -1/3 & 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 251 & 405 & 235 \\ 405 & 891 & 405 \\ 235 & 405 & 251 \end{bmatrix} //$$

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Orthogonal Matrix: A square matrix P is called an orthogonal matrix iff $P^{-1} = P^T$.

Theorem: A real square matrix is orthogonally diagonalizable if and only if it is symmetric.

Spectral Theorem: Any $n \times n$ symmetric matrix has the following properties:

1. A has n real eigenvalues, counting the algebraic multiplicities.
2. The dimension of eigenspace of each eigenvalue λ equals the algebraic multiplicities.
3. Eigenvectors corresponding to distinct eigenvalues are orthogonal.
4. A is orthogonally diagonalizable.

Working Rule: Let A be an $n \times n$ symmetric matrix.

1. Find all eigenvalues of A and determine their algebraic multiplicities.
2. Find the corresponding eigenvectors: for distinct eigenvalues take unit eigenvectors, and for repeated eigenvalues obtain a set of linearly independent eigenvectors and orthonormalize them using the Gram–Schmidt process.
3. Form the orthogonal matrix P, using the **orthonormal** eigenvectors as columns.
4. Then $P^{-1}AP = P^TAP = D$, where D is a diagonal matrix containing the eigenvalues of A.

Note: An $n \times n$ matrix P is orthogonal if and only if its column vectors form an orthogonal set.

II Orthogonally diagonalize the following matrices.

1. $A = \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$.

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C.E. is,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where,

$$S_1 = 0$$

$$S_2 = -4 - 4 - 4 = -12$$

$$S_3 = 16$$

Thus,

$$\lambda^3 - 12\lambda - 16 = 0$$

$\therefore \lambda = 4, -2, -2$

for $\lambda = 4$

$$A - \lambda I = \begin{bmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{bmatrix}$$

$R_1 \to R_1 + 2R_2$

$R_3 \to R_3 - R_2$

$$\approx \begin{bmatrix} 0 & -6 & 6 \\ 2 & -4 & 2 \\ 0 & 6 & -6 \end{bmatrix}$$

$R_3 \to R_3 + R_1$

$$\approx \begin{bmatrix} 0 & -6 & 6 \\ 2 & -4 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n-r = 1$ free variable

Let,

$z = k$

$\therefore y = k$

$\therefore x = k$

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$$\therefore V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for $\lambda = -2$

$$A - \lambda I = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$$\approx \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n-r = 2$ free va

Let, $z = k_1$, $y = k_2$

$\therefore x = -k_1 - k_2$

$$\therefore X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k_1 - k_2 \\ k_2 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\therefore V_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad V_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Since,

V_2 & V_3 are orthogonal to V_1 but not with each other.

Thus,

Apply Gram-Schmidt method

i.e.

Let, $u_2 = V_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

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$$u_3 = V_3 - \frac{\langle V_3\ u_2 \rangle}{\langle u_2\ u_2 \rangle} \cdot u_2$$

$$= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

Thus.

$$V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

now,

$$\hat{V}_1 = \frac{V_1}{\|V_1\|} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

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$$\hat{u}_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix},$$

$$\hat{u}_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

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So,

$$P = [\ \hat{v}_1 \ \|\ \hat{u}_2 \ \|\ \hat{u}_3\ ]$$

$$= \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 2/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

And,

$$P^{-1} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \end{bmatrix} = P^T$$

Thus

$$D = P^{-1}AP = P^T AP$$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \end{bmatrix} \cdot \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} //$$

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2. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

C. E. is,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where:

$$S_1 = 0$$

$$S_2 = -1 - 1 - 1 = -3$$

$$S_3 = 2$$

So

$$\lambda^3 - 3\lambda - 2 = 0$$

$\therefore \lambda = 2, -1, -1$

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for $\lambda = 2$

$$A - \lambda I = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + 2R_2$$
$$R_3 \rightarrow R_3 - R_2$$

$$\approx \begin{bmatrix} 0 & -3 & 3 \\ 1 & -2 & 1 \\ 0 & 3 & -3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1$$

$$\approx \begin{bmatrix} 0 & -3 & 3 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore n - r = 1$ free var

let,

$z = k$

$\therefore y = k$

$\therefore x = k$

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$\therefore V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

for $\lambda = -1$

$$A - \lambda I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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Let, $z = k_1$, $y = k_2$

$\therefore x = -k_2 - k_1$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -k_1 - k_2 \\ k_2 \\ k_2 \end{bmatrix} = k_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$\therefore V_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $V_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

Since.

V_2 & V_3 are orthogonal to V_1 but not with each other.

So Apply G-S method

Let,

$$u_2 = V_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

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$$u_3 = V_3 - \frac{\langle V_3 u_2 \rangle}{\langle u_2 u_2 \rangle} \cdot u_2$$

$$= \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{1}{2} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

So,

$$\hat{v}_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \hat{u}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \hat{u}_3 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

$$\therefore \quad P = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

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And

$$P^{-1} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \end{bmatrix} = P^T$$

Thus

$$D = P^{-1}AP = P^T A P$$

$$= \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \times$$

$$\begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} //$$

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3. $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$.

Cof. is,

$S_1 = 17$

$S_2 = 29 + 29 + 32 = 90$

$S_3 = 144$

So,

$\lambda^3 - 17\lambda^2 + 90\lambda - 144 = 0$

$\therefore \lambda = 3, 8, 6$ Since all root are distinct, no G-S method is needed

for $\lambda = 3$

$$A - \lambda I = \begin{bmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

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$R_1 \rightarrow R_1 + 3R_3$

$R_2 \rightarrow R_2 - 2R_3$

$$\approx \begin{bmatrix} 0 & -5 & 5 \\ 0 & 5 & -5 \\ -1 & -1 & 2 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 5 & -5 \\ -1 & -1 & 2 \end{bmatrix}$$

$\therefore n - r = 1$ free var

Let.

$z = k$

$\therefore y = k$

$\therefore x = k$

$$\therefore V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for $\lambda = 8$

$$A - \lambda I = \begin{bmatrix} -2 & -2 & -1 \\ -2 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & 0 & 5 \\ 0 & 0 & 5 \\ -1 & -1 & -3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 5 \\ -1 & -1 & -3 \end{bmatrix}$$

$\therefore$ $n - r = 1$ free variable

Let.

$y = k$

$\therefore$ $z = 0$

$\therefore$ $x = -k$

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$$\therefore V_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

for $\lambda = 6$

$$A - \lambda I = \begin{bmatrix} 0 & -2 & -1 \\ -2 & 0 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$R_2 \to R_2 - 2R_3$

$$\approx \begin{bmatrix} 0 & -2 & -1 \\ 0 & 2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$$

$R_1 \to R_1 + R_2$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 1 \\ -1 & -1 & -1 \end{bmatrix}$$

$\therefore$ $n - r = 1$ free vari-

Let, $z = k$

$\therefore y^2 - k/2$

$\therefore x = -\frac{k}{2}$

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$$\therefore V_3 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

Thus.

$$V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad V_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad V_3 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

So,

$$\hat{V}_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \hat{V}_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \hat{V}_3 = \begin{bmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$P = [\hat{V}_1 \,||\, \hat{V}_2 \,||\, \hat{V}_3]$$

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$$= \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}$$

Also

$$P^{-1} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix} = P^T$$

Thus,

$$D = P^{-1}AP$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 6 \end{bmatrix} \quad //$$

4. $A=\begin{bmatrix}2 & 2 & 4\\ 2 & 5 & 8\\ 4 & 8 & 17\end{bmatrix}$

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C.E. is,

$S_1 = 24$

$S_2 = 21+18+6 = 45$

$S_3 = 22$

Thus,

$\lambda^3 - 24\lambda^2 + 45\lambda - 22 = 0$

$\therefore \lambda = 22, 1, 1$

for $\lambda = 22$

$A - \lambda I = \begin{bmatrix}-20 & 2 & 4\\ 2 & -17 & 8\\ 4 & 8 & -5\end{bmatrix}$

$R_1 \to R_1 + 10R_2$

$R_3 \to R_3 - 2R_2$

$\approx \begin{bmatrix}0 & -168 & 84\\ 2 & -17 & 8\\ 0 & 42 & -21\end{bmatrix}$

$R_1 \to R_1 + 4R_3$

$\approx \begin{bmatrix}0 & 0 & 0\\ 2 & -17 & 8\\ 0 & 42 & -21\end{bmatrix}$

Let,

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$z = k$

$\therefore y = \frac{k}{2}$

$\therefore x = k/4$

$\therefore\ V_1 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$

for $\lambda = 1$

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 4 & 8 & 16 \end{bmatrix}$$

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$$R_2 \to R_2 - 2R_1$$

$$R_3 \to R_3 - 4R_1$$

$$\approx \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore\ n - r = 2$ free variable

Let,

$z = k_2,\ y = k_1$

$\therefore\ x = -2k_1 - 4k_2$

So

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2k_1 - 4k_2 \\ k_1 \\ k_2 \end{bmatrix}$$

$$= k_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore\ V_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix},\ V_3 = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$

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$\therefore v_1 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$

Since v_2 & v_3 are orthogonal to v_1 but not to each other.

So, By G-S method

Let,

$$u_2 = v_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

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$$u_3 = v_3 - \frac{\langle v_3 \, u_2 \rangle}{\langle u_2 \, u_2 \rangle} \cdot u_2$$

$$= \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} - \frac{8}{5} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} -4 \\ -8 \\ 5 \end{bmatrix}$$

So.

$$\hat{v}_1 = \begin{bmatrix} 1/\sqrt{21} \\ 2/\sqrt{21} \\ 4/\sqrt{21} \end{bmatrix},$$

$$\hat{u}_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}$$

$$\hat{u}_3 = \begin{bmatrix} -4/\sqrt{105} \\ -8/\sqrt{105} \\ 5/\sqrt{105} \end{bmatrix}$$

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$$\therefore P = \begin{bmatrix} 1/\sqrt{21} & -2/\sqrt{5} & -4/\sqrt{105} \\ 2/\sqrt{21} & 1/\sqrt{5} & -8/\sqrt{105} \\ 4/\sqrt{21} & 0 & 5/\sqrt{105} \end{bmatrix}$$

Also,

$$P^{-1} = \begin{bmatrix} 1/\sqrt{21} & 2/\sqrt{21} & 4/\sqrt{21} \\ -2/\sqrt{5} & 1/\sqrt{5} & 0 \\ -4/\sqrt{105} & -8/\sqrt{105} & 5/\sqrt{105} \end{bmatrix} = P^T$$

Thus.

$$D = P^T A P = P^{-1} A P$$

$$= \begin{bmatrix} 22 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} //$$

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QUADRATIC FORMS:
A function $Q: R^n \to R$ is called a quadratic form if $Q(X) = X^T AX = \sum_{i,j=1}^{n} A_{ij} X_i X_j$ where A is a symmetric matrix.

Canonical Form:
Let A be the matrix of the quadratic form Q and let λ_1, λ_2 and λ_3 are eigen values of A and v_1, v_2 and v_3 are corresponding orthonormal eigen vectors of A and $P = [v_1\ v_2\ v_3]$ be an orthogonal matrix then $X^T P^{-1} APX = \lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$ is called canonical form or principal axes form of the quadratic form.

Nature of the Quadratic form: A quadratic form Q is
1. Positive Definite if $Q(X) > 0\ \forall\ x \neq 0 \Rightarrow \lambda_i > 0$ for all i.
2. Negative Definite if $Q(X) < 0\ \forall\ x \neq 0 \Rightarrow \lambda_i < 0$.
3. Positive Semidefinite if $Q(X) \geq 0\ \forall\ x \Rightarrow \lambda_i \geq 0$.
4. Negative Semidefinite if $Q(X) \leq 0\ \forall\ x \Rightarrow \lambda_i \leq 0$.
5. Indefinite if $Q(X)$ has both positive and negative values.

Index of a Quadratic form: The number of positive terms in its canonical form is called the index of a quadratic form.

Rank of the Quadratic form: The number of non-zero terms in its canonical form is called rank of the quadratic form.

Signature of the Quadratic form: Signature of the quadratic form is the difference of the positive and negative terms in the canonical form.

III. Determine the modal matrix which reduces the following quadratic forms to the canonical forms. Hence find nature, rank, signature and index of the quadratic forms:

1. $x^2 + 3y^2 + 3z^2 - 2yz$. —— I

→ Standard form,

$f = ax^2 + by^2 + cz^2 + 2dxy + 2eyz + 2fxz$ —— II

Compare eqn I & II

$a = 1,\ b = 3,\ c = 3,\ d = 0,\ e = -1,\ f = 0$

∴ The matrix of the Quadratic form.

$$A = \begin{bmatrix} a & d & f \\ d & b & e \\ f & e & c \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

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C.E.

$S_1 = 7$

$S_2 = 8 + 3 + 3 = 14$

$S_3 = 8$

Thus. $\lambda^3-7\lambda^2+14\lambda-8=0$

$\therefore \lambda=4,2,1$

for $\lambda=4$

$$A-\lambda I=\begin{bmatrix}-3&0&0\\0&-1&-1\\0&-1&-1\end{bmatrix}$$

$$\approx\begin{bmatrix}-3&0&0\\0&-1&-1\\0&0&0\end{bmatrix}$$

$\therefore n-r=1$ free v

Let

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$z=k$

$\therefore y=-k$

$\therefore x=0$

$$\therefore P_1=\begin{bmatrix}0\\-1\\1\end{bmatrix}$$

for $\lambda=2$

$$A-\lambda I=\begin{bmatrix}-1&0&0\\0&1&-1\\0&-1&1\end{bmatrix}$$

$$\approx\begin{bmatrix}-1&0&0\\0&1&-1\\0&0&0\end{bmatrix}$$

$n-r=1$

Let

$z=k$

$\therefore y=k$ $\quad\therefore P_2=\begin{bmatrix}0\\1\\1\end{bmatrix}$

$\therefore x=0$

for $\lambda = 1$

$$A - \lambda I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

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$$R_2 \longrightarrow R_2 + 2R_3$$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

$n - r = 1$

Let,

$x = K$

$\therefore y = 0$

$\therefore z = 0$

$$\therefore P_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

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So

for $\lambda_1 = 4$, $P_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

for $\lambda_2 = 2$, $P_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

for $\lambda_3 = 1$, $P_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Since, all eigen values are distinct they are orthogonal to each other.

now,

we just have to make it orthonormal,

So,

$$\hat{P}_1 = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad \hat{P}_2 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad \hat{P}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Thus Modal Matrix

$$P = [\hat{P}_1 \| \hat{P}_2 \| \hat{P}_3]$$

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$$= \begin{bmatrix} 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

&

$$P^{-1} = P^T$$

$\Rightarrow P$ is orthogonal matrix

So,

$$D = P^{-1} A P = P^T A P$$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The Canonical form is

$$\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$$

$$= 4x^2 + 2y^2 + z^2$$

Nature: Since all eigen value > 0

$\therefore$ positive definite

Index: No. of +ve eigen value $= 3$

Rank : 3

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Signature: diff b/w no. of +ve and -ve eigen value

$\Rightarrow 3 - 0 = 3$

2. $3x^2+5y^2+3z^2-2xy+2xz-2yz$.

↳ standard form.

$$ax^2+by^2+cz^2+2dxy+2eyz+2fxz$$

$\therefore a=3,\ b=5,\ c=3,\ d=-1,\ e=-1,\ f=1$

Quadratic Matrix

$$A=\begin{bmatrix}3 & -1 & 1\\ -1 & 5 & -1\\ 1 & -1 & 3\end{bmatrix}$$

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C.E.

$S_1=11$

$S_2=14+8+14=36$

$S_3=36$

Thus,

$$\lambda^3-11\lambda^2+36\lambda-36=0$$

$\therefore \lambda=6,3,2$

for $\lambda=6$

$$A-\lambda I=\begin{bmatrix}-3 & -1 & 1\\ -1 & -1 & -1\\ 1 & -1 & -3\end{bmatrix}$$

$R_2\to 3R_2-R_1$

$R_3\to 3R_3+R_1$

$$\approx\begin{bmatrix}-3 & -1 & 1\\ 0 & -2 & -4\\ 0 & -4 & -8\end{bmatrix}$$

$$\approx\begin{bmatrix}-3 & -1 & 1\\ 0 & -2 & -4\\ 0 & 0 & 0\end{bmatrix}$$

$n - r = 1$

Let, $z = K$

$\therefore y = -2K$

$\therefore x = K$

$$\therefore P_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

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for $\lambda = 3$

$$A - \lambda I = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\approx \begin{bmatrix} -1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$n - r = 1$

Let

$z = K$

$\therefore y = K$

$\therefore x = K$

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$$\therefore P_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

for $\lambda = 2$,

$$A - \lambda I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$R_2 \to R_2 + R_1$

$R_3 \to R_3 - R_1$

$$\approx \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $z = K$

$\therefore y = 0$

$\therefore x = -K$

$$\therefore P_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

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Thus.

$$\hat{P}_1 = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \hat{P}_2 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \hat{P}_3 = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Modal Matrix,

$$P = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \\ -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$

&

$P^{-1} = P^T$

$= P$ is orthogonal Matrix

Now, $D = P^{-1}AP = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Canonical form: $6x^2 + 3y^2 + 2z^2$

Nature: positive Definite

Index: 3

Rank: 3

Signature: $3 - 0 = 3$ //

3. $x^2+5y^2+z^2+2xy+6xz+2yz$.

↳ standard form

$ax^2+by^2+cz^2+2dxy+2eyz+2fxz$

So. $a=1, b=5, c=1, d=1, e=1, f=3$

Quadratic Matrix

$$A=\begin{bmatrix}1 & 1 & 3\\ 1 & 5 & 1\\ 3 & 1 & 1\end{bmatrix}$$

C. E.

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$S_1=7$

$S_2=4-8+4=0$

$S_3=-36$

Thus,

$\lambda^3-7\lambda^2+36=0$

$\therefore \lambda=-2, 6, 3$

for $\lambda=-2$

$$A-\lambda I=\begin{bmatrix}3 & 1 & 3\\ 1 & 8 & 1\\ 3 & 1 & 3\end{bmatrix}$$

$R_1 \to R_1-3R_2$

$R_3 \to R_3-3R_2$

$$\approx\begin{bmatrix}0 & -23 & 0\\ 1 & 8 & 1\\ 0 & -23 & 0\end{bmatrix}$$

$$\approx\begin{bmatrix}0 & -23 & 0\\ 1 & 8 & 1\\ 0 & 0 & 0\end{bmatrix}$$

$x - x = 1$

Let, $z = k$ $\quad \therefore P_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$y = 0$

$\therefore x = -k$

for $\lambda = 6$

$$A - \lambda I = \begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

$R_1 \to R_1 + 5R_2$

$R_3 \to R_3 - 3R_2$

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$$\approx \begin{bmatrix} 0 & -4 & 8 \\ 1 & -1 & 1 \\ 0 & 4 & -8 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & -4 & 8 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Let.

$z = k$ $\quad \therefore P_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$\therefore y = 2k$

$\therefore x = k$

for $\lambda = 3$

$$A - \lambda I = \begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix}$$

$R_1 \to R_1 + 2R_2$

$R_3 \to R_3 - 3R_2$

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$$\approx \begin{bmatrix} 0 & 5 & 5 \\ 1 & 2 & 1 \\ 0 & -5 & -5 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & 5 & 5 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $z = K$

$\therefore y = -K$

$\therefore x = K$

$$P_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Thus.

$$\hat{P}_1 = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \hat{P}_2 = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \hat{P}_3 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Modal Matrix -

$$P = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

&

$$P^{-1} = P^T$$

$\therefore$ P is orthogonal mon

So,

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

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$\therefore$ Canonical form: $-2x^2 + 6y^2 + 3z^2$

Rank $= 3$

Index $= 2$ (Two +ve eigen value i.e. 6 & 3)

Nature = Indefinite

signature $= 2 - 1 = 1$ //

4. $2xy + 2xz - 2yz$.

Standard form

$\hookrightarrow ax^2 + by^2 + cz^2 + 2dxy + 2eyz + 2fxz$

$\therefore a = 0, b = 0, c = 0, d = 1, e = -1, f = 1$

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

C.E.

$S_1 = 0$

$S_2 = -1 - 1 - 1 = -3$

$S_3 = -2$

Thus

$\lambda^3 - 3\lambda + 2 = 0$

$\therefore \lambda = -2, 1, 1$

for $\lambda = -2$

$$A - \lambda I = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$R_1 \rightarrow R_1 - 2R_2$

$R_3 \rightarrow R_3 - R_2$

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$$\approx \begin{bmatrix} 0 & -3 & 3 \\ 1 & 2 & -1 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & -3 & 3 \\ 1 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Let, $z = k$ $\therefore P_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$\therefore y = k$ $\therefore x = -k$

for $\lambda = 1$

$$A - \lambda I = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$R_2 \to R_2 + R_1$

$R_3 \to R_3 + R_1$

$$\simeq \begin{bmatrix} -1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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Let

$z = k_1, \quad y = k_2$

$\therefore x = k_1 + k_2$

So, $x = \begin{bmatrix} k_1 + k_2 \\ k_2 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Thus

$P_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad P_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Since P_2 & P_3 are orthogonal to P_1 but not with each other

So. use G-S method

Let,

$$u_2 = P_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$u_3 = P_3 - \frac{\langle P_3 u_2 \rangle}{\langle u_2 u_2 \rangle} \cdot u_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

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Thus.

$$\hat{P}_1 = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \hat{u}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \hat{u}_3 = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

Modal matrix

$$P = \begin{bmatrix} -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

&

$$P^{-1} = P^T$$

$\therefore$ P is orthogonal mat

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So.

$$D = P^{-1} A P$$

$$= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ Canonical form $= -2x^2 + y^2 + z^2$

$\therefore$ nature = Indefinite

Index = 2

Rank = 3

Signature = 2 - 1 = 1

5. $3x^2+3y^2+3z^2+2xy+2xz-2yz$.

↳ Standard form,

$ax^2+by^2+cz^2+2dxy+2eyz+2fxz$

$\therefore a=3,\ b=3,\ c=3,\ d=1,\ e=-1,\ f=1$

$$A=\begin{bmatrix}3 & 1 & 1\\ 1 & 3 & -1\\ 1 & -1 & 3\end{bmatrix}$$

C.E.

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$S_1=9$

$S_2=8+8+8=24$

$S_3=16$

Thus.

$\lambda^3-9\lambda^2+24\lambda-16=0$

$\therefore \lambda=1,4,4$

for $\lambda=1$

$$A-\lambda I=\begin{bmatrix}2 & 1 & 1\\ 1 & 2 & -1\\ 1 & -1 & 2\end{bmatrix}$$

$R_1\to R_1-2R_2$

$R_3\to R_3-R_2$

$$\approx\begin{bmatrix}0 & -3 & 3\\ 1 & 2 & -1\\ 0 & -3 & 3\end{bmatrix}$$

$$\approx\begin{bmatrix}0 & -3 & 3\\ 1 & 2 & -1\\ 0 & 0 & 0\end{bmatrix}$$

Let $z=K$

$\therefore y=K\quad \therefore x=-K\quad \therefore P_1=\begin{bmatrix}-1\\ 1\\ 1\end{bmatrix}$

for $\lambda = 4$

$$A - \lambda I = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$R_1 \to R_1 + R_2$

$R_3 \to R_3 - R_2$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $z = k_2, \; y = k_1$

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$\therefore \; x = k_1 + k_2$

Then

$$X = \begin{bmatrix} k_1 + k_2 \\ k_1 \\ k_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore \; P_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \; P_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Apply G-S method

Let $$u_2 = P_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$u_3 = P_3 - \frac{\langle P_3 \; u_2 \rangle}{\langle u_2 \; u_2 \rangle} \cdot u_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Thus, $\hat{P}_1 = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$, $\hat{u}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$, $\hat{u}_3 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$

So,

$$P = \begin{bmatrix} -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}$$

&

$$P^{-1} = P^T$$

$\Rightarrow$ P is orthogonal matrix

Thus $D = P^{-1}AP$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

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$\therefore$ Canonical form: $x^2 + 4y^2 + 4z^2$

Rank $= 3$

Nature $=$ positive definite

Index $= 3$

Signature $= 3 - 0 = 3$

6. $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$

Standard form,

$$ax^2 + by^2 + cz^2 + 2dxy + 2eyz + 2fxz = 0$$

So,

$$a = 6,\ b = 3,\ c = 3,\ d = -2,\ e = -1,\ f = 2$$

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

C.E.

$$S_1 = 12$$

$$S_2 = 8 + 14 + 14 = 36$$

$$S_3 = 32$$

Thus

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\therefore \lambda = 8, 2, 2$$

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for $\lambda = 8$

$$A - \lambda I = \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}$$

$$R_1 \to R_1 + R_3$$

$$R_2 \to R_2 + R_3$$

$$\simeq \begin{bmatrix} 0 & -3 & -3 \\ 0 & -6 & -6 \\ 2 & -1 & -5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & -6 & -6 \\ 2 & -1 & -5 \end{bmatrix}$$

Let $z = K$

$\therefore y = -K,\ x = 2K$ $\quad \therefore P_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

for $\lambda = 9$

$$A - \lambda I = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0 & 0 & 0 \\ -2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Let, $z = k_1$, $y = k_2$

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$$\therefore x = \frac{k_2 - k_1}{2}$$

Thus.

$$X = \begin{bmatrix} \frac{k_2 - k_1}{2} \\ k_2 \\ k_1 \end{bmatrix} = \frac{k_2}{2} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \frac{k_1}{2} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$\therefore P_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad P_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \quad P_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Not orthogonal

Apply G-S Method

Let,

$$V_2 = P_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$V_3 = P_3 - \frac{\langle P_3 \, V_2 \rangle}{\langle V_2 \, V_2 \rangle} \cdot V_2$$

$$= \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} -4 \\ 2 \\ 10 \end{bmatrix}$$

$$= \frac{2}{5}\begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$$

$$\therefore \hat{P}_1 = \begin{bmatrix} 2/\sqrt{6} \\ -1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \hat{P}_2 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

$$\hat{P}_3 = \begin{bmatrix} -2/\sqrt{30} \\ 1/\sqrt{30} \\ 5/\sqrt{30} \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 2/\sqrt{6} & 1/\sqrt{5} & -2/\sqrt{30} \\ -1/\sqrt{6} & 2/\sqrt{5} & 1/\sqrt{30} \\ 1/\sqrt{6} & 0 & 5/\sqrt{30} \end{bmatrix}$$

&

$P^T = P^{-1}$

$\therefore$ P is orthogonal matrix

So

$D = P^T A P$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$\therefore$ canonical form $= 8x^2 + 2y^2 + 2z^2$

Rank = 3

Index = 3

Nature : positive Definite

Signature : 3 − 0 = 3 //

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ALL THE BEST